



Indian Institute of Information Technology, Surat
Department of Computer Science & Engineering
End Semester Examination, 3rd Sem, 2025-26
Programming for Problem Solving (CS 303)

Timing: 10:00 AM to 01:00 PM

Date: 06/12/2025

Max Marks: 100

Note: Attempt all parts of a question at one place.

1. (a) Write the answer of the following

1. Predict output:

```
s = "Engineering"  
print(s[3:8:2])
```

10x2=20

2. Identify the error (if any) and correct it:

```
t = (10)  
t.append(20)
```

3. Write a list comprehension to extract only **unique vowels** from the string "Programming".

4. What will be printed?

```
d = {'a':10, 'b':10, 'c':20}  
print(len(set(d.values())))
```

5. Write a lambda function to compute $(x^2 + y^2)$.

6. Predict output: `import numpy as np, a = np.array([1,2,3,4]), print(a[a%2==0])`

7. Name the technique : `a + .5`

8. Write a Pandas command to remove rows having missing values.

9. Define data cleaning with one example.

10. Predict output:

```
print(list(zip([10,20], ['x','y','z'])))
```

2. (a) Write a Python program that: Accepts a sentence, Removes all spaces, Reverses the sentence using **slicing** and Displays the frequency of each character using a **dictionary**

2x10=20

(b) Create a Python program using: **List comprehension** to generate squares of numbers divisible by 3 from 1 to 50, **Lambda + filter** to retain results > 200

3 (a) Predict the output and explain the logic:

10x2=20

```
import numpy as np  
a = np.array([[1,2,3],[4,5,6]])  
print(a[a>3])
```

(b) Using Python and NumPy:

Create a 3x3 matrix of ones
Multiply it with any constant using **Broadcasting**
Replace values greater than 2 using `np.where()`

10x2=20

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Write any two:

- (a) Write a Python + NumPy program to:
Merge two arrays horizontally

Split the combined array into three equal parts

- (b) Write Pandas code to:
Create DataFrame from dictionary
Add a column "Total" using vector arithmetic
Remove missing rows
Sort records descending by "Total"

- (c) Explain with charts
Any **two data visualization techniques**
Their role in **Exploratory Data Analysis (EDA)**

10x2=20

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Write any two:

- (a) A CSV file contains student scores with missing values.

Explain and demonstrate:

Reading file into DataFrame,
Cleaning missing data,
Adding a calculated column: "Total".

- (b) Define **Exploratory Data Analysis (EDA)** and explain:

Any **two visualization techniques**
Steps to detect outliers and data anomalies

- (c) Explain the following using examples:

Lambda function, Boolean Indexing, Broadcasting
Exploratory Data Analysis (EDA)